

What's Your 1st Guess?

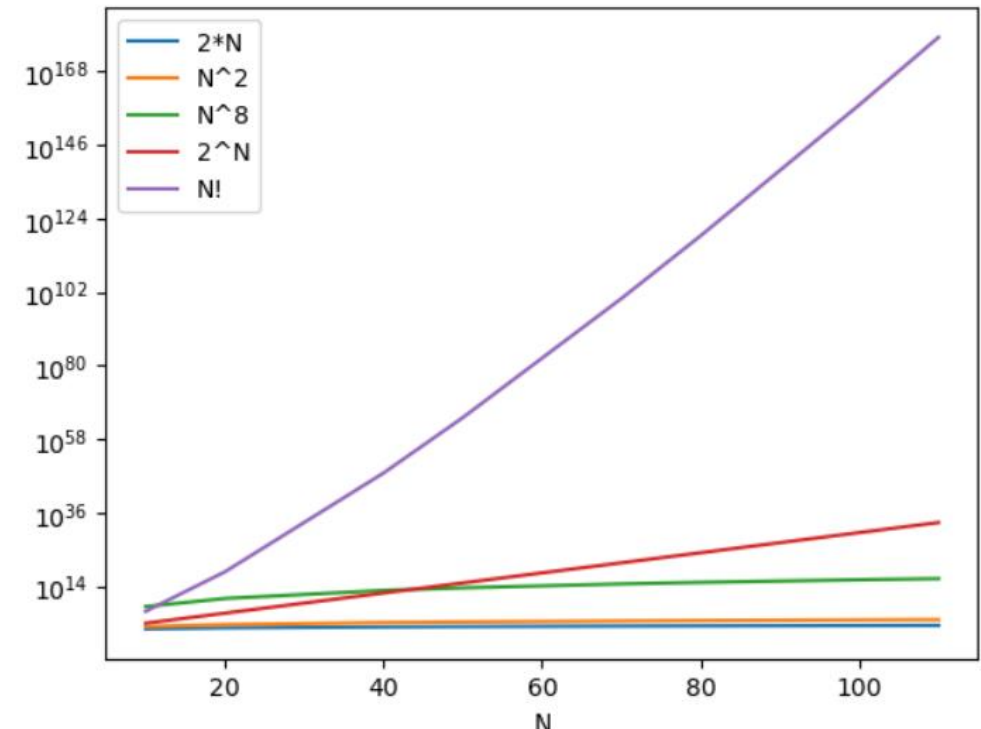
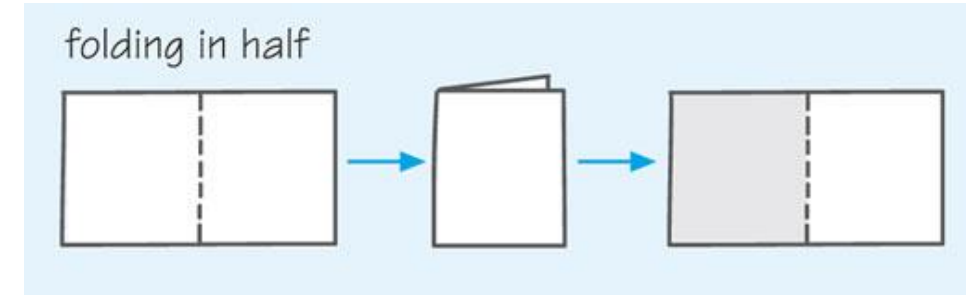
- ▶ Assume we have a piece of infinitely large thin paper

- Thickness: 0.1mm (10^{-4}m)
- Earth to the Moon: $3.8 \times 10^8\text{m}$

How many times we need to fold the paper, so it reaches the Moon from the Earth?

- A. 42 B. 420
C. 4200 D. none of these

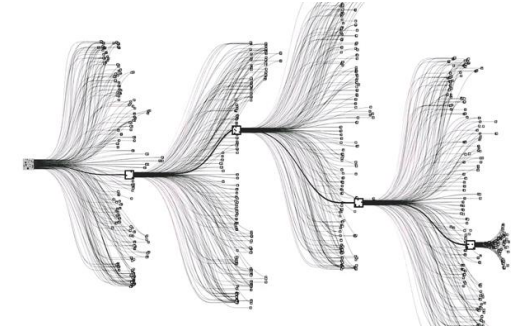
- ▶ Is there any other problem of similar characteristic?



Course Info – Lecture Schedule

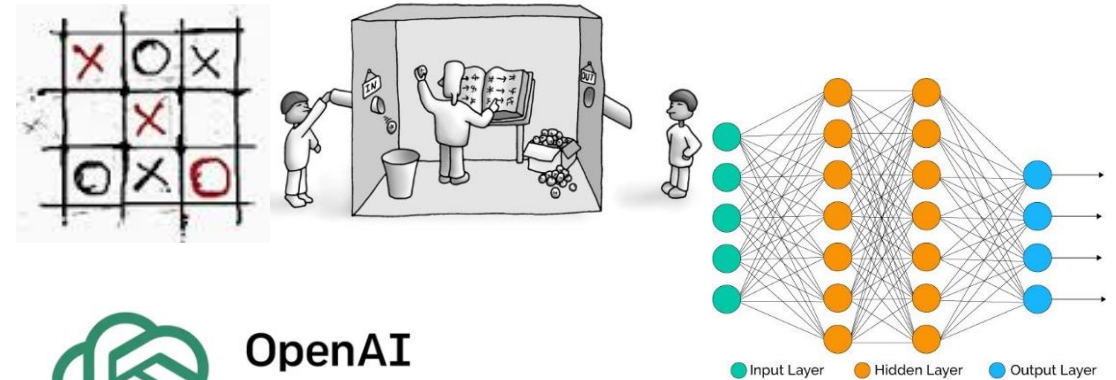
► Part I: Fundamental Issues

- AI milestones, AI experiments, brief history, developments



► Part II: Machine learning

- Artificial Neural Networks (ANN)
- Other techniques (Regression, Decision Tree)



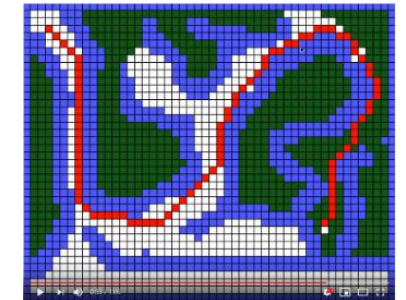
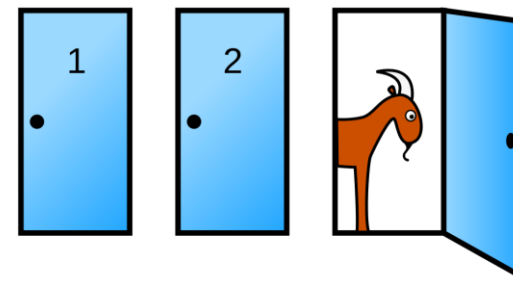
► Part III: Intelligent techniques

- Bayes Rule, KRR
- Natural Language Processing, LLMs



► Part IV: Tree search techniques

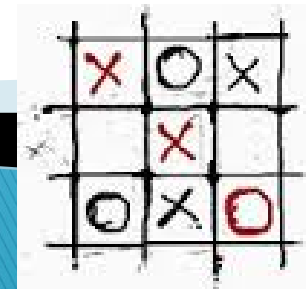
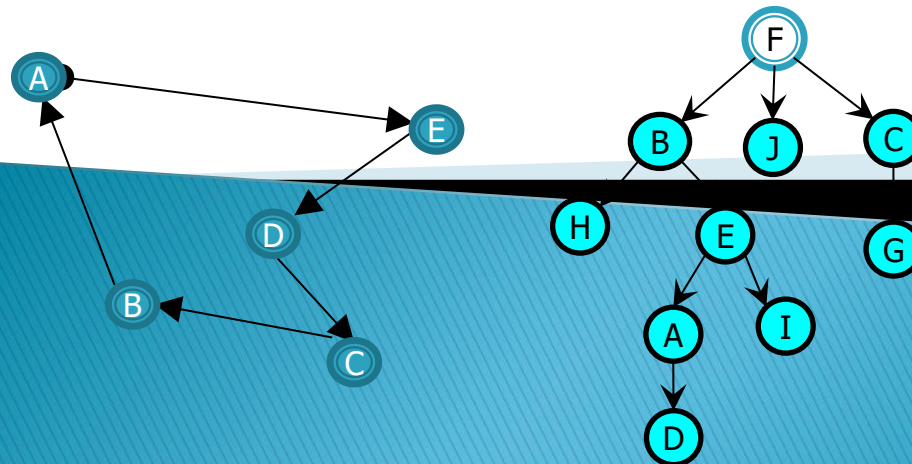
- Combinatorial explosion
- Depth First, Breadth First, A*
- Game Playing (MiniMax, Alpha-beta pruning)



COMP1008 - Fundamentals of AI

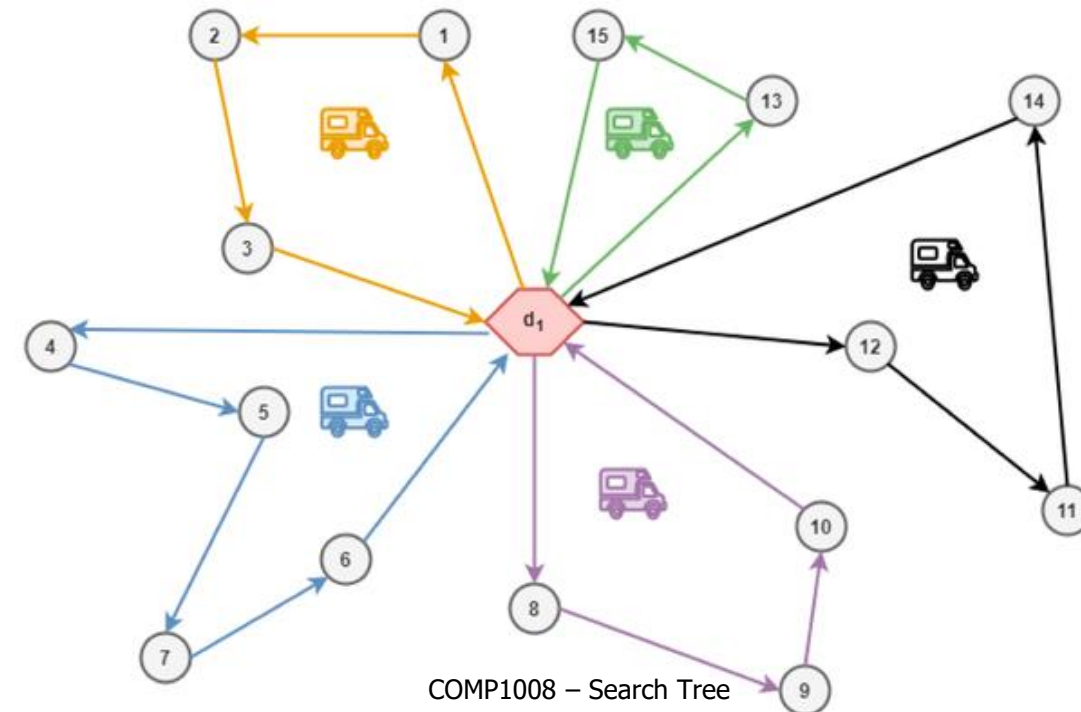
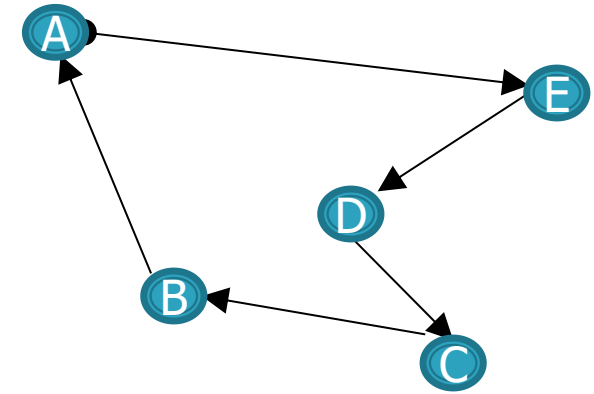
Search Tree

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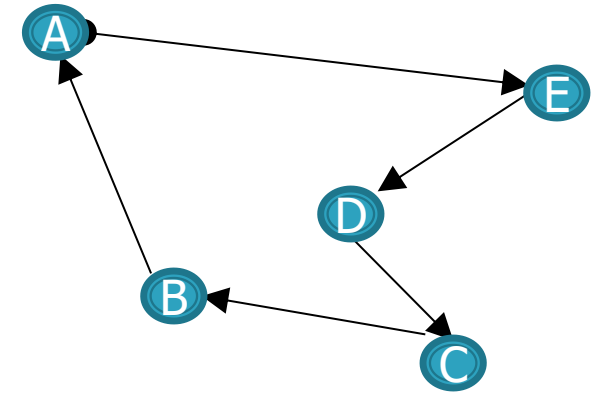


The Travelling Salesman Problem

- ▶ A salesperson must visit n cities
 - Start at any city and must finish at that same city
 - Must visit each city only **once**
- ▶ Solving TSP: to find the **minimum distance / cost** solution
 - Given a set of n cities and distances / costs between them
 - The number of possible routes is ?



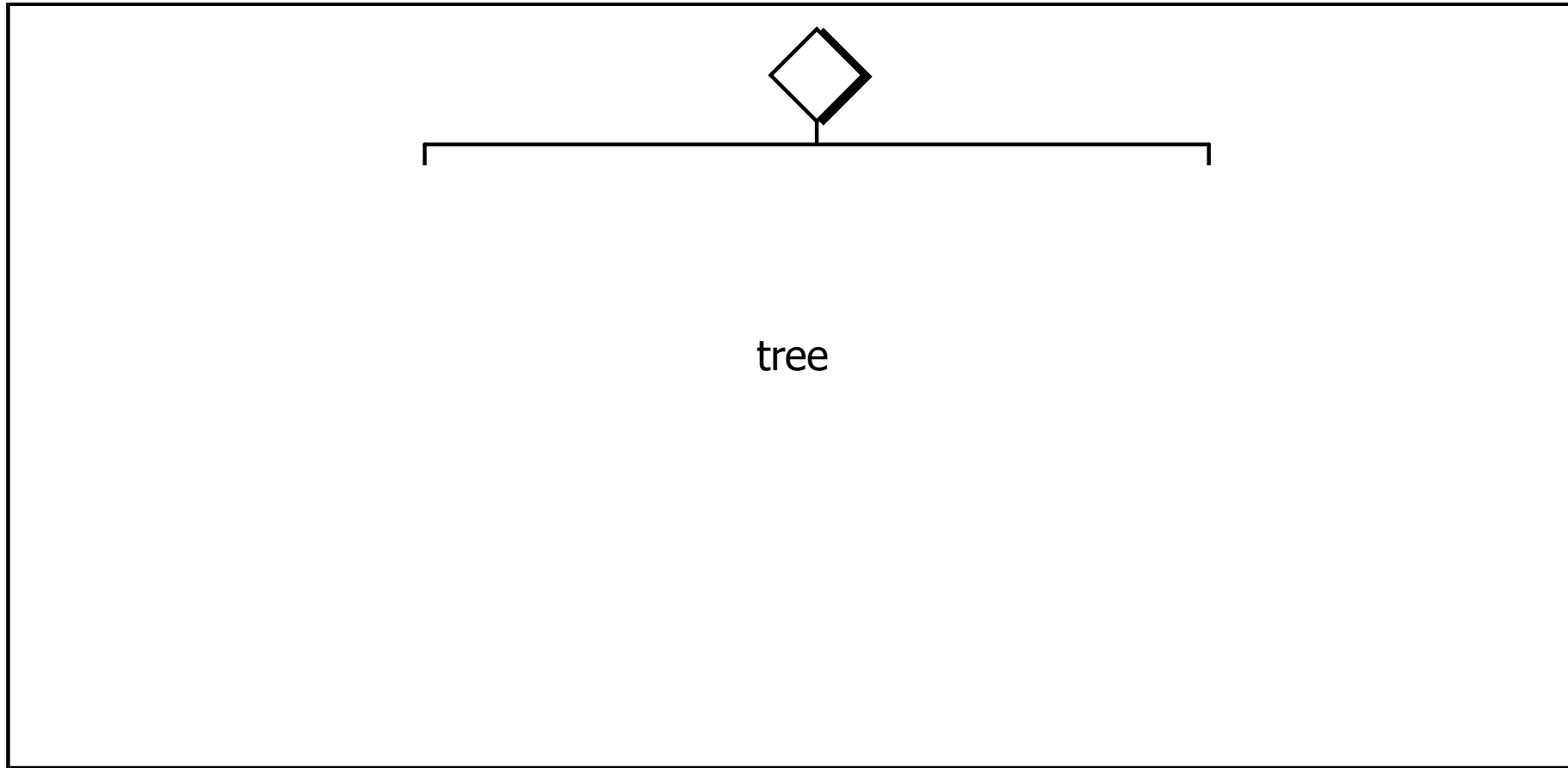
The Travelling Salesman Problem



- ▶ A 10 city TSP has 181,000 possible solutions
- ▶ A 20 city TSP has 10,000,000,000,000,000 possible solutions
- ▶ A 50 city TSP has 100,000 possible solutions ($1.52 * 10^{64}$)
- ▶ There are 1,000,000,000,000,000,000,000 liters of water on the planet *
- ▶ A 10GHz computer might do 10^9 tours per second
- ▶ Running since start of universe, it would only have done 10^{26} tours
- ▶ **Combinatorial Explosion Problem:** One of the major unsolved theoretical problems in Computer Science

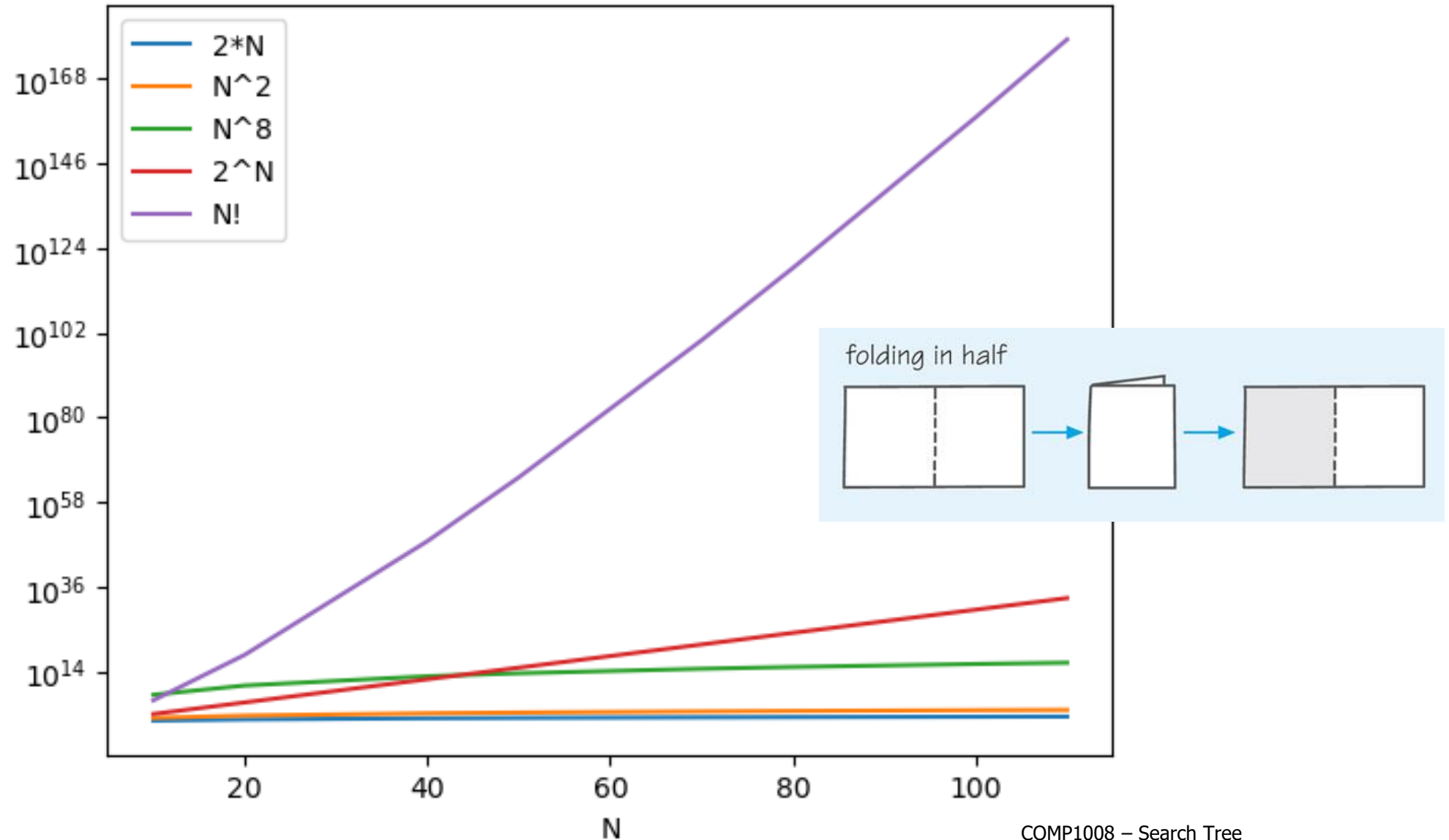
* Mchalewicz, Z, Evolutionary Algorithms for Constrained Optimization Problems, CEC 2000 (Tutorial)

Combinatorial Explosion



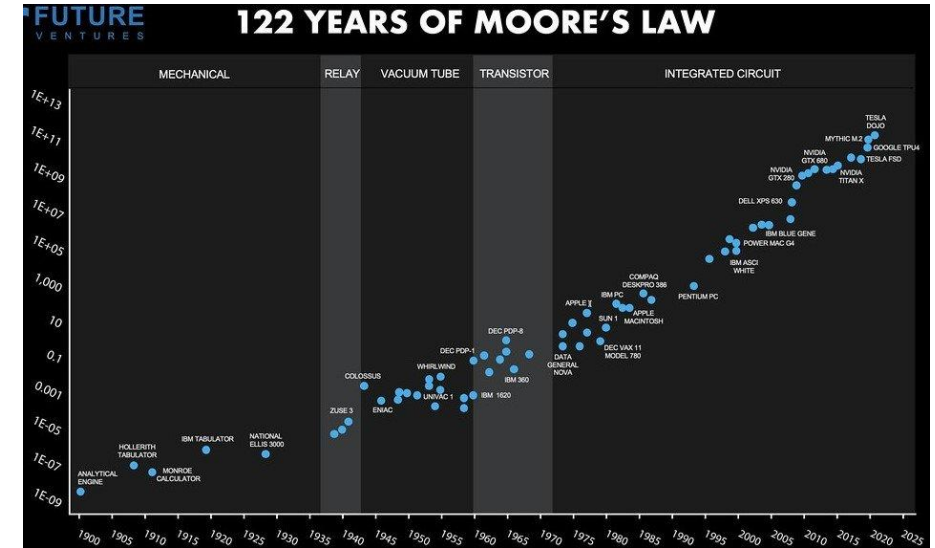
The feature where the number of solutions for a problem grows **exponentially** with the problem size

Combinatorial Explosion



Combinatorial Explosion – Examples

- ▶ Question: any other examples?



Search Tree - Problem Search Space

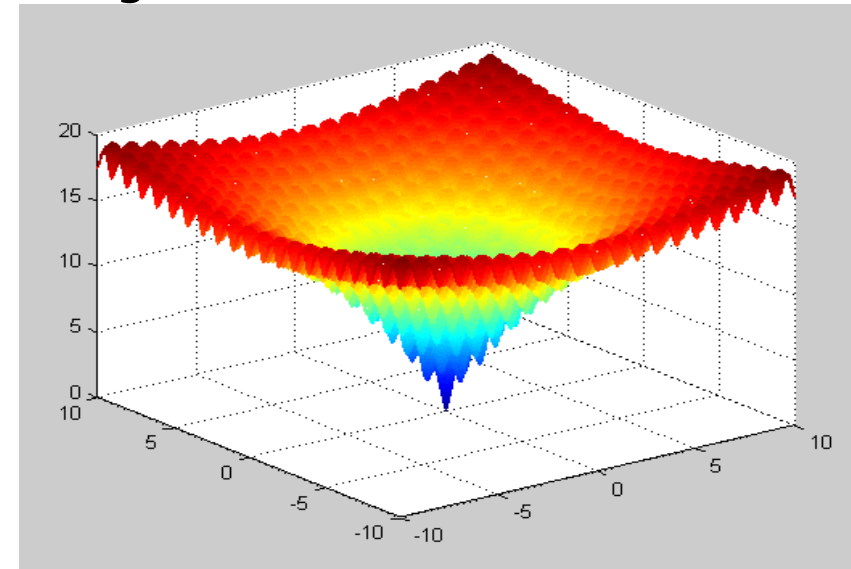
- ▶ The concept of **Search**
 - Important in science and engineering
 - A lot of problems can be seen as a **search** for “the right answer”

Quadratic equation in standard form

$$ax^2 + bx + c = 0$$

Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$F(\vec{x}) = -20 \cdot \exp\left(-0.2 \sqrt{\frac{1}{n} \cdot \sum_{i=1}^n x_i^2}\right) - \exp\left(\frac{1}{n} \sum_{i=1}^n \cos(2\pi \cdot x_i)\right) + 20 + e$$
$$x_i \in [-30, 30]$$

Search Tree - Problem Search Space

- ▶ The concept of **Search Space**
 - Set of **all possible solutions** to a problem
- ▶ Search algorithms
 - Input: a problem
 - Return: a solution to the problem

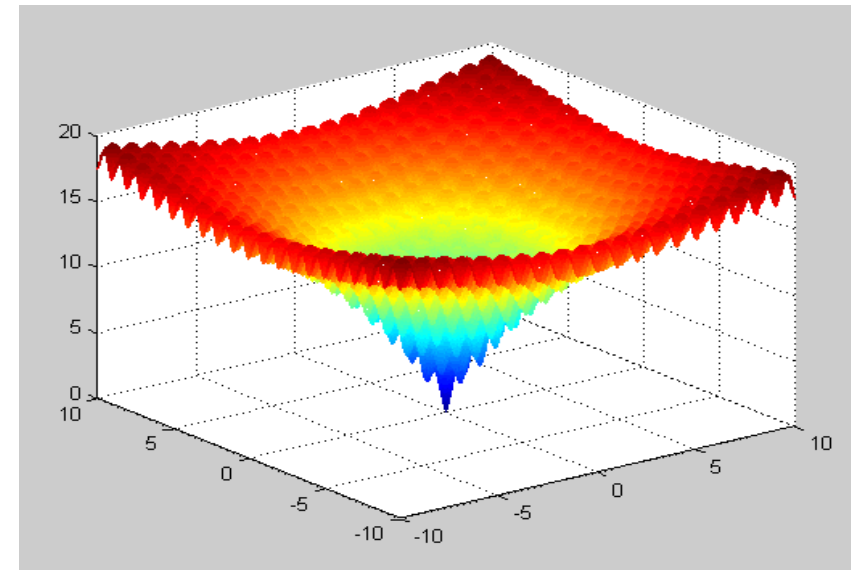


COMP1008

CS Course

Search techniques

COMP2001 – AI Methods

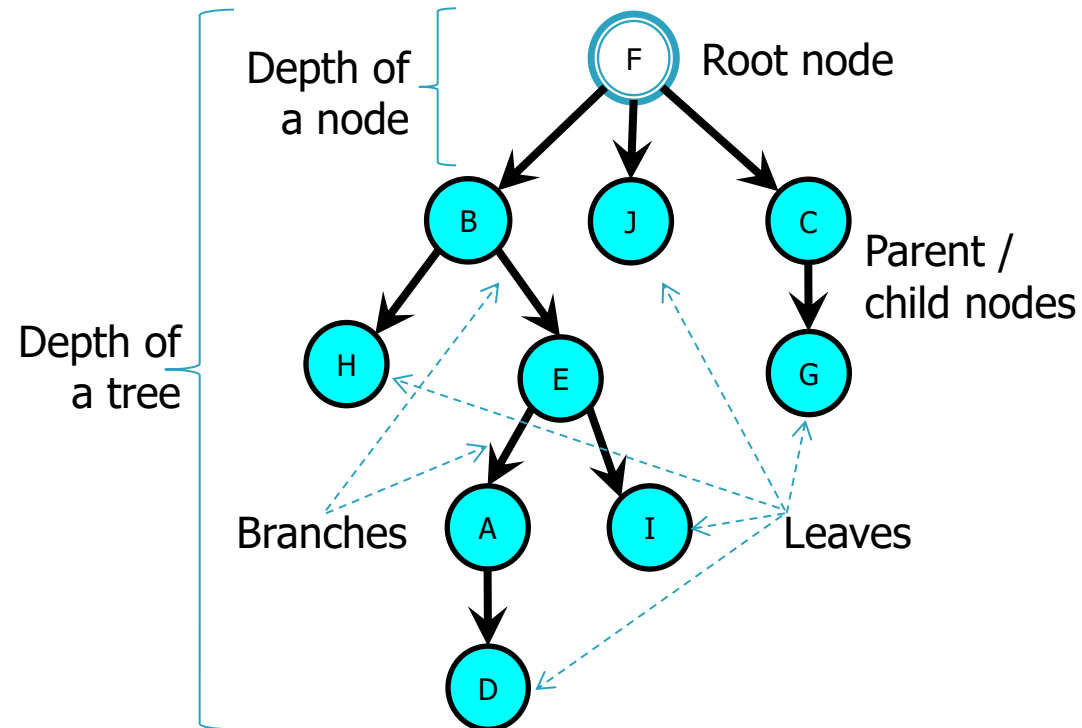


$$F(\vec{x}) = -20 \cdot \exp\left(-0.2 \sqrt{\frac{1}{n} \cdot \sum_{i=1}^n x_i^2}\right) - \exp\left(\frac{1}{n} \sum_{i=1}^n \cos(2\pi \cdot x_i)\right) + 20 + e$$
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Search Tree - Problem Search Space

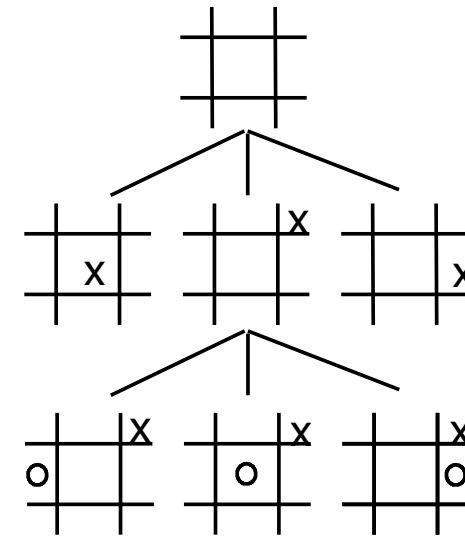
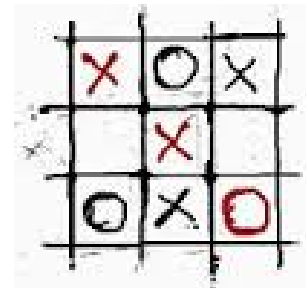
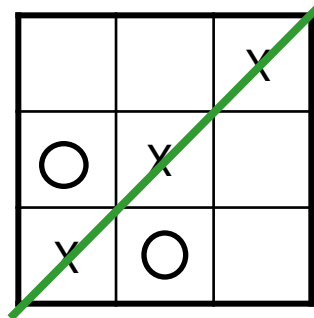


- ▶ The concept of **Search Space**
 - Set of **all possible solutions** to a problem



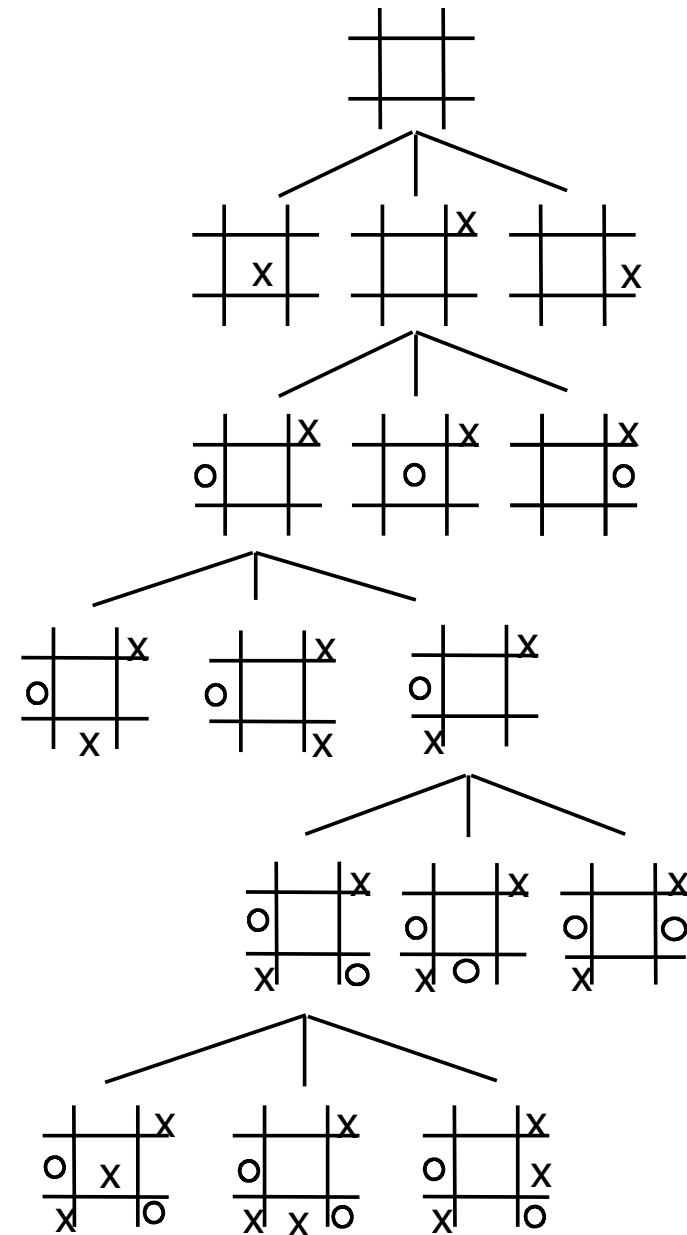
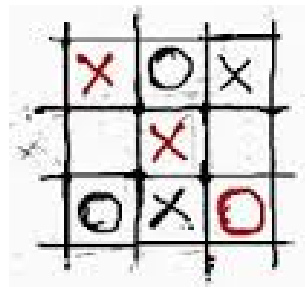
Search Tree – Define a Problem

- ▶ Define a search problem using a **search tree**
 - **States** - nodes
 - Possible states of the problem
 - **Search Space** – all nodes in the tree
 - The set of all states reachable from the initial state
 - **Operator** - branches
 - A set of actions that move one state to another



Search Tree – Define a Problem

- ▶ Define a search problem using a search tree
 - **Neighbourhood**
 - All possible states reachable from a given state
 - **Goal Test**
 - A test to a state to tell if the search reached a **state that solves the problem**
 - **Path Cost**
 - How much it **costs to take a particular path**



Search Tree – Define a Problem: Example

5	4	
6	1	8
7	3	2

Initial State

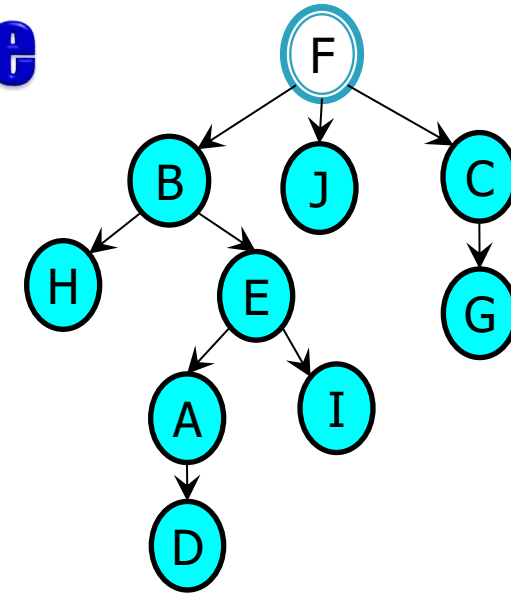
1	4	7
2	5	8
3	6	

Goal State

[Online examples](#)

Search Tree – Define a Problem: Example

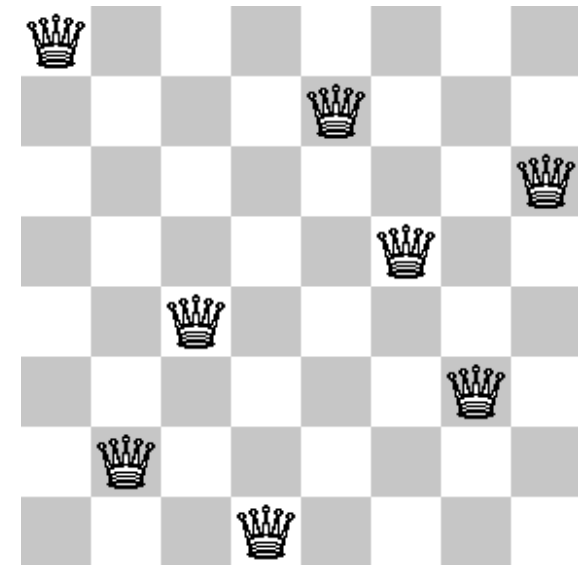
- ▶ States
 - A **board** of the eight tiles at their locations
 - It is also useful to include the **blank**
- ▶ Operator
 - A tile moves left, right, up or down
 - The blank **moves** left, right, up or down
- ▶ Goal Test
 - The current state matches one of the aimed **goal states**
- ▶ Path Cost
 - Each move of a tile costs 1
 - Each move of the blank **costs** 1



5	4	
6	1	8
7	3	2

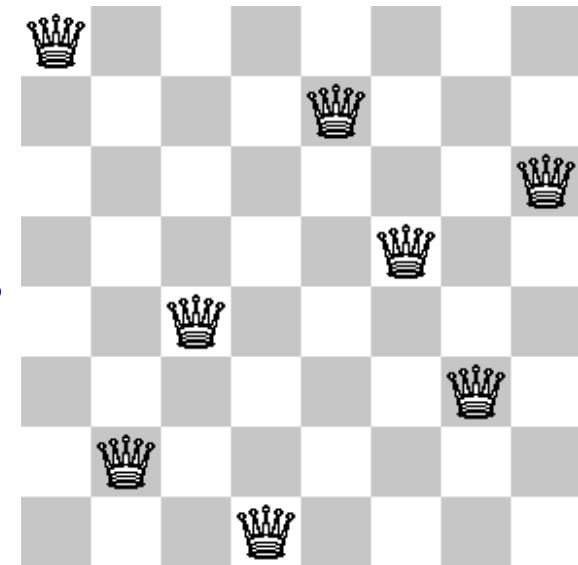
Search Tree – Define a Problem: Past Exam Question

- ▶ Define the states and operator of the 8-Queen problem. (4 marks)
- ▶ Comment on the size of the search tree in relation to combinatorial explosion. (3 marks)
The 8-Queen problem aims to place 8 chess queens on a chess board so as no two queens attack each other.
- ▶ Initial state - an empty board
- ▶ Operator
 - add a queen into a cell on the board
- ▶ States
 - any arrangement of 0 to 8 queens on the board
- ▶ Goal state
 - a valid arrangement of 8 queens on board
- ▶ **Question:** Other ways to define the 8-queen problem?



Search Tree – Define a Problem: Past Exam Question

- ▶ 1st level: 1 root node (empty board)
 - ▶ 2nd level: 64 nodes
 - ▶ 3rd level: 63 nodes for each of the node on the 2nd level
 - ▶ ...
-
- ▶ Improve search techniques:
 1. Define the problem into **smaller** search tree: [model](#)
 2. Improve **search efficiency** through the tree: [techniques](#)

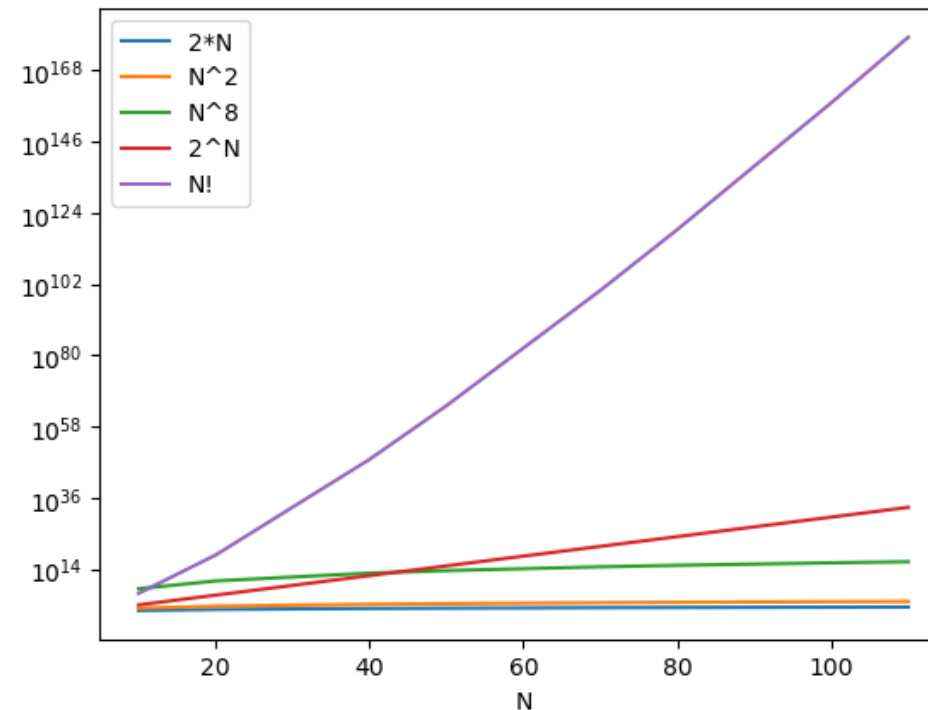


Search Tree – Issues

- ▶ Tree size
 - **Branching factor**: average no. of branches of all nodes in a tree
 - Example: $b = 2$ (binary tree), depth d

d	nodes at d , 2^d	total nodes
0	1	1
1	2	3
2	4	7
3	8	15
4	16	31
5	32	63
6	64	127

Exponentially - Combinatorial explosion



Search Tree – Summary

- ▶ Concept of **Combinatorial Explosion**
 - Definition, Examples
- ▶ Define a problem into a search problem
 - **Search tree**: states vs. nodes, operator vs. branching factor, solutions vs. path
 - Examples

